MTAT.07.003 Cryptology II
Spring 2012 / Exercise session ?? / Example Solution

Exercise (CCA security for commitments). *Consider an opening oracle $\mathcal{O}$ against a commitment scheme* $(\mathsf{Gen}, \mathsf{Com}, \mathsf{Open})$ *that is defined in the following way*

$$\begin{array}{l} \mathcal{O}_{\mathsf{pk}}(c) \\ \left[\begin{array}{l} d_* \xleftarrow[u]{} \{d : \mathsf{Open}_{\mathsf{pk}}(c, d) \neq \bot\} \\ \textbf{\textit{return}}\ d_* \end{array}\right. \end{array} .$$

Let $\mathcal{L}$ denote the list of inputs submitted to $\mathcal{O}_{\mathsf{pk}}$. Then the commitment scheme is (t, q, ε)-binding against opening oracle the advantage of any t-time adversary $\mathcal{A}$ in the following game

$$\begin{array}{l} \mathcal{G} \\ \left[\begin{array}{l} \mathsf{pk} \leftarrow \mathsf{Gen} \\ c, d_0, d_1 \leftarrow \mathcal{A}^{\mathcal{O}_{\mathsf{pk}}(\cdot)}(\mathsf{pk}) \\ \mathsf{if}\ \mathsf{Open}_{\mathsf{pk}}(c, d_0) = \bot\ \mathsf{then}\ \textbf{\textit{return}}\ 0 \\ \mathsf{if}\ \mathsf{Open}_{\mathsf{pk}}(c, d_1) = \bot\ \mathsf{then}\ \textbf{\textit{return}}\ 0 \\ \mathsf{if}\ c \in \mathcal{L} \vee |\mathcal{L}| > q\ \mathsf{then}\ \textbf{\textit{return}}\ 0 \\ \textbf{\textit{return}}\ \mathsf{Open}_{\mathsf{pk}}(c, d_0) \neq \mathsf{Open}_{\mathsf{pk}}(c, d_1) \end{array}\right. \end{array}$$

is bounded by ε, i.e., $\Pr\left[\mathcal{G}^{\mathcal{A}} = 1\right] \leq \varepsilon$. Assume that there exists a τ-time adversary $\mathcal{B}$ that has a large advantage $\mathsf{Adv}(\mathcal{B})$ *against the following game pairs*

$$\begin{array}{l} \mathcal{Q}_0 \\ \left[\begin{array}{l} \mathsf{pk} \leftarrow \mathsf{Gen} \\ m_0, m_1 \leftarrow \mathcal{B}(\mathsf{pk}) \\ c, d \leftarrow \mathsf{Com}_{\mathsf{pk}}(m_0) \\ c_* \leftarrow \mathcal{B}(c) \\ d_* \leftarrow \mathcal{B}(d) \\ \textbf{\textit{return}}\ \mathsf{Open}_{\mathsf{pk}}(c_*, d_*) = m_0 \end{array}\right. \end{array} \qquad \begin{array}{l} \mathcal{Q}_1 \\ \left[\begin{array}{l} \mathsf{pk} \leftarrow \mathsf{Gen} \\ m_0, m_1 \leftarrow \mathcal{B}(\mathsf{pk}) \\ c, d \leftarrow \mathsf{Com}_{\mathsf{pk}}(m_1) \\ c_* \leftarrow \mathcal{B}(c) \\ d_* \leftarrow \mathcal{B}(d) \\ \textbf{\textit{return}}\ \mathsf{Open}_{\mathsf{pk}}(c_*, d_*) = m_1 \end{array}\right. \end{array}$$

where

$$\mathsf{Adv}(\mathcal{B}) = \frac{1}{2} \cdot \Pr\left[\mathcal{Q}_0^{\mathcal{B}} = 1\right] + \frac{1}{2} \cdot \Pr\left[\mathcal{Q}_1^{\mathcal{B}} = 1\right] - \frac{1}{2} \quad .$$

Show that then the commitment scheme cannot be $(t, 1, \varepsilon)$-binding

Solution. For simplicitly analyse the case where each commitment c has only two valid openings d_0 and d_1 and both lead to different messages.